Brandon J. Chen

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PROFESSIONAL EXPERIENCE

Project Engineer / 2022-Present
Lea+Elliott, Inc.
Assist AGT/APM system planning initiatives, including design, project management, construction oversight

- **Seattle-Tacoma International Airport** - Satellite Transit System Controls Renewal and Replacement
- **Singapore Changi International Airport** - Terminal 5 Expansion
- **Los Angeles International Airport** - Landside Access Modernization Program (SkyLink)
- **Fort Lauderdale** Airport-Seaport-Convention Center Connector

Design Engineer / 2018-2022
RSE, Inc. / Belmont, CA

- **Caltrain, signal system maintenance**: Updated signal diagram drawings systemwide; Worked with client in conjunction with upcoming track electrification, positive train control, and other track construction projects; Trained other engineers and delegated work as needed
- **Santa Clara Valley Transportation Authority, light rail system rehabilitation**: Processed survey point clouds; Created topography for rehabilitation candidate locations
- **Caltrain, Grade Crossing Hazard Analysis**: Analyzed on-ground conditions at all grade crossings, calculated potential risks of anticipated incidents, and proposed curated safety improvements; Managed interns
- **Los Angeles Metro, West Santa Ana Branch Extension**: Processed field survey data from 40 crossings into preliminary computer models for track design phase; Coordinated with project manager and contractors on the schedule, standards, and deliverables
- **BART, PreScience track geometry car development**
- **Caltrain, Positive Train Control** (collision prevention system): Project witness and field test engineer

Design Engineer Intern / 2016
Filtran LLC (Filtration Group) / Des Plaines, IL
Designed transmission fluid filters for major automotive manufacturers; Overhauled QA equipment designs

Design Intern / 2015
Segal Design Institute / Evanston, IL
Co-led Segal floor outlet cover replacement; Designed and built prototype infant surgical warming pad

EDUCATION

Northwestern University, B.S. Design and Manufacturing Engineering / 2013-2017

Courses of Interest: Computer Integrated Manufacturing, Energy Sustainability, Human-Centered Product Design, Intellectual Property & Innovation, Interdisciplinary Design Projects, Operations Research

Activities: 3D4E (3D Printing for Everyone) club co-founder / 2016-2017

Capstone: "Bolt" Breadfruit Peeler / 2016
Designed motorized peeler that reduced breadfruit-to-flour conversion workload for Jamaican farmers

SKILLS

Computer-Aided Design: AutoCAD, Civil3D, MicroStation, Revit; Creo Pro/E, Siemens NX, SolidWorks
Survey: Leica Cyclone, ReCap, TopoDOT
Planning: Project controls / project management, design review, construction oversight, alignment design, survey analysis

Product Design: Human-computer interaction, industrial design, life-cycle analysis, mechanical design, UX research; technical writing, product journey mapping
Manufacturing & Statistics: Design of experiments, GD&T, lean manufacturing, materials selection
Software: Adobe Illustrator, InDesign, Photoshop, CorelDraw; ArcGIS, Microsoft Office